INAIO 2025 Stage 1 Results

February 5, 2025

Scoring Process

The scoring process was designed to ensure fairness and account for varying difficulty levels in the questions. Below are the key points regarding the score computation

- **Bonus Points Allocation**: A total of 20 bonus points were awarded based on the difficulty of questions for large language models (LLMs). Refer to the answer key and points table at the end of this document for the point allocation.

- **Question Exclusion**: Question 10 from the short problems section was excluded from scoring due to an error in the problem matrix.

- **Ambiguous Questions**: For questions where ambiguity existed, or where multiple reasonable answers were possible due to rounding issues, all reasonable answers were evaluated positively. Refer to the answer key for specific details.

- **Real-Value Rounding**: Following multiple student requests and considering that this is the first edition of the test, real-valued numerical answers were rounded down before evaluation.

- **No Negative Marking**: There was no negative marking for incorrect answers.

Selection Criteria

The primary criteria for student selection was to balance performance across both short foundational questions and long ML-themed questions while also allow students who performed exceptionally well either of these to progress further. Performance of LLMs served as a benchmark for cut-off calibration.

Average LLM Performance

LLM	Short QA Score	Long QA Score
Gemini	13	30
ChatGPT (Free)	13.5	32
ChatGPT (Paid)	13	43

Cut-off Scores

Cut-off thresholds were calibrated to ensure the cut-off for grades 9 and 10 aligns with the performance of a free LLM. The exact cut-offs are listed below.

Grade Group	Short QA Cut-off	Long QA Cut-off.	Global Short QA Cut-off.	Global Long QA Cut-off.
8 and below	12.5	25	16.5	50
9-10	13	31	16.5	50
11-12	14	38	16.5	50

Selection Rule

Students were selected only if they satisfy one of the following criteria:

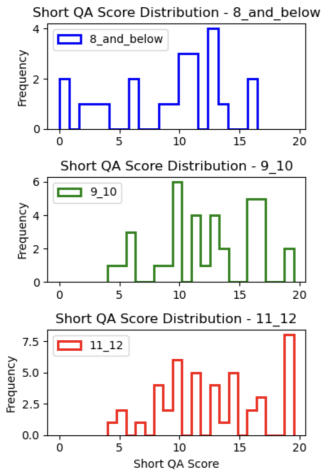
- `(Short QA Score` $\geq$ `Grade Group Short QA Cut-off) AND (Long QA Score` $\geq$ `Grade Group Long QA Cut-off)`
- `Short QA Score` $\geq$ `Global Short QA Cut-off`
- `Long QA Score` $\geq$ `Global Long QA Cut-off`

Revaluation Process

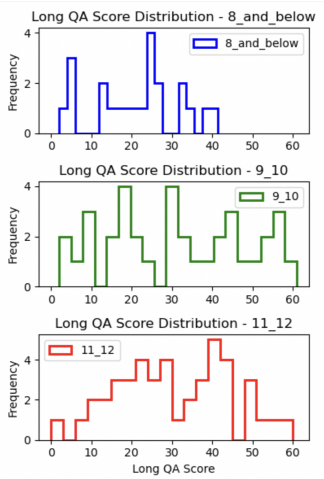
Students who believe their scores do not match expectations after reviewing the answer key may request revaluation by reaching out to `inaio@acmindia.org` to submit your request before Feb 10. The final decision regarding revaluation will rest with INAIO.

Score Distribution

Below we show the grade-wise score distributions for both short and long QA scores.



(a) Short QA Score Distribution



(b) Long QA Score Distribution

Figure 1: Score Distributions by Grade Group for Short and Long QA

Student Disqualification

Responses were analyzed to detect patterns indicating the likelihood of student collusion. In cases where we could establish a high confidence of students cheating, they were disqualified regardless of their scores. As a result, 13 students from four schools have been disqualified. While it is possible that there have been other cases of unethical behavior, we were not able to establish it with high confidence. The upcoming Stage 2 test will be proctored and both the telemetric and response data will be analyzed to ensure fairness for all the students.

Section Name	Question	Question ID	Answer Key	Additional Comments	Original Points	Bonus Points
Short Problems	Wizardly Wason Safety Check	Q1	[1, 2, 7]		2	0
Short Problems	RoboChef's Taste Test	Q2	86		2	0
Short Problems	Perfectly Priced Park Ticket	Q3	55		2	0
Short Problems	Mystery Balls and Balances	Q4	(a) 3, (b) Not Possible, (c) 5		2	1
Short Problems	Fraud Filters	Q5	Maximize Precision at a Coverage Bound		2	1
Short Problems	Revenue RollerCoaster Function	Q6	Option 2		2	1
Short Problems	Study-Play Balance for Success	Q7	390		2	0
Short Problems	Party Planning Puzzle	Q8	37		2	0
Short Problems	Water Balloon Toss	Q9	22		2	1.5
Short Problems	Similarity Sleuth	Q10	Option 1	Due to an error in the question data, we have nullified this question	2	-2
Alumni Donor Classifier	Confusion Matrix Calculation	Q1(a)	17		1	1
Alumni Donor Classifier	Confusion Matrix Calculation	Q1(b)	11		1	1
Alumni Donor Classifier	Confusion Matrix Calculation	Q1(c)	5		1	0

Alumni Donor Classifier	New Classifier (AB) Definition	Q2(a)	86 or 0.86	Due to ambiguity in the question, both 86 and 0.86 are acceptable with matching up to 2nd decimal for the latter choice	2	1
Alumni Donor Classifier	New Classifier (AB) Definition	Q2(b)	84 or 0.84	Due to ambiguity in the question, both 84 and 0.84 are acceptable with matching up to 2nd decimal for the latter choice	2	0
Alumni Donor Classifier	ROC and Class Distribution	Q3(a)	89 or 0.89	Due to ambiguity in the question, both 89 and 0.89 are acceptable with matching up to 2nd decimal for the latter choice	2	1
Alumni Donor Classifier	ROC and Class Distribution	Q3(b)	The curve remains unchanged		2	1
Alumni Donor Classifier	ROC and Class Distribution	Q3(c)	The curve flips across the diagonal		2	1
Alumni Donor Classifier	Mis-classification Cost	Q4	19		2	2
Matrix Magic	Performing Convolution	Q1	43		3	1.5
Matrix Magic	Stride and Padding	Q2	49		2	0
Matrix Magic	Blurring	Q3	Option 1		2	0
Matrix Magic	Brightening	Q4	Option 4		2	0
Matrix Magic	Sharpening	Q5	Option 1		2	0

Matrix Magic	Edge Detection	Q6	Option 2		2	0
Matrix Magic	No Change	Q7	Option 2		2	0
From Words to Vectors	Predicting Missing Word Embeddings	Q1	[2, 2]		2	0
From Words to Vectors	Semantic Difference between Word Embeddings	Q2	Royalty		2	0
From Words to Vectors	Maximizing Sentence Incoherence	Q3(a)	20		2	0
From Words to Vectors	Maximizing Sentence Incoherence	Q3(b)	89		5	1
From Words to Vectors	Maximizing Sentence Incoherence	Q3(c)	70		4	1
PageRank	Applying PageRank	Q1(a)	9 or 10	Due to rounding issues, both 9 and 10 are acceptable	2	1
PageRank	Applying PageRank	Q1(b)	32		1	1.5
PageRank	Applying PageRank	Q1(c)	User C		1	1
PageRank	Convergence Properties	Q2(a)	Convergence becomes slower and final values become more sensitive to the link structure (represented by W)		2	0
PageRank	Convergence Properties	Q2(b)	The principal eigenvector of the transition matrix T		2	0

PageRank	Sink Components	Q3(a)	Users C, E, F		1	0
PageRank	Sink Components	Q3(b)	Sink components will absorb most of the PageRank, leaving less for nodes outside		2	0
PageRank	Influence Maximization	Q4(a)	Both Options 1 and 3		2	1.5
PageRank	Influence Maximization	Q4(b)	The group minimizes outgoing links to users outside the group		2	1